

Requirement Specification Document

E-Commerce Order Management Database System

Week 1 – Requirement Analysis and Customer Database Module

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Project Title	Online Food Delivery System
Week	Week 1

1. Introduction

1.1 Purpose

The purpose of this document is to analyze and specify the business requirements of the **Online Food Delivery System**. This system enables customers to browse restaurant menus, place food orders, make online payments, and track deliveries in real time. It provides an efficient platform for customers, restaurants, delivery partners, and administrators.

1.2 Project Scope

The system allows customers to register, browse restaurants, search for food items, place orders, make secure payments, and monitor delivery status. Restaurants can manage menus and orders, while delivery partners update delivery progress. The system uses a relational database to store customer, restaurant, menu, order, and payment information.

1.3 Intended Users

- Customers – Browse menus, place food orders, and track deliveries.
- Restaurant Managers – Manage menu items and customer orders.
- Delivery Partners – Accept and deliver customer orders.
- Administrators – Manage restaurants, customers, and reports

2. Business Requirements

The system must satisfy the following business requirements

- Maintain customer information securely.
- Maintain restaurant and menu details.
- Allow customers to place food orders online.
- Process online and cash-on-delivery payments.
- Track order status in real time.
- Assign delivery partners to customer orders.
- Generate daily, weekly, and monthly sales reports

3. Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement Description
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FR-1	The system shall allow customers to register using their name, email, phone number, and address
FR-2	The system shall allow restaurants to add, update, and remove food items.
FR-3	The system shall allow customers to browse restaurants and menus.
FR-4	The system shall allow customers to place orders for one or more food items.
FR-5	The system shall automatically calculate the total bill including taxes and delivery charges.
FR-6	The system shall record payment details using UPI, cards, wallets, or Cash on Delivery.
FR-7	The system shall track order status (Order Placed, Preparing, Picked Up, Out for Delivery, Delivered, Cancelled).
FR-8	The system shall generate sales and order reports for administrators.

4. Non-Functional Requirements

- **Performance:** The system should display restaurant menus within 2 seconds.
- **Security:** Customer passwords and payment information must be encrypted.
- **Reliability:** Orders should not be duplicated or lost.
- **Availability:** The system should be available 24×7.
- **Usability:** The application should provide an easy-to-use interface.
- **Scalability:** The database should support thousands of restaurants and customers.

5. Customer Database Module – Data Requirements

Since this week's focus includes the Customer Database Module, the following entity and attributes have been identified:

Attribute	Data Type	Description
customer id	INT (Primary Key)	Unique customer ID
full name	VARCHAR(100)	Customer name
email	VARCHAR(100)	Customer email
phone number	VARCHAR(10)	Customer contact number
address	VARCHAR(255)	Delivery address
registration date	DATE	Account creation date

6. Assumptions and Constraints

- The system will use MySQL, PostgreSQL, or Oracle Database.
- Only registered customers can place food orders.
- Every order must belong to one customer and one restaurant.
- Internet connectivity is required.
- Payment gateways must be available for online payments..

7. Conclusion

This document captures the initial business and functional requirements for the E-Commerce Order Management. The **Online Food Delivery System** provides an efficient platform for customers to order food online while helping restaurants manage orders and delivery operations. This document defines the initial business, functional, and non-functional requirements for the project. It serves as the foundation for designing the ER diagram, database schema, and implementation in the upcoming development phases.